Simultaneous valuation conditions and explicit quartics

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Abstract

We give a valuation argument that replaces several multiplicative local tests by one polynomial expression. In the elliptic construction over $\mathbb{Q}(\sqrt{2})$, powers already computed for the formal logarithm provide the required terms without additional high-degree equations. Sixteen local tests of three different orders are expressed by five or six such polynomials. The complete coefficient tables specify quartics in 301 and 351 total variables. Under the stated arithmetic premises, sufficiently large members of the 301-variable family define $\mathbb{Z}$ in $\mathbb{Q}$, and an explicit member of the 351-variable family does so. The argument preserves the formal-depth requirements, the scalar normalization, and the numerical sufficient parameter for the explicit construction.

1 Local tests with different valuation classes

Let v be a discrete valuation with value group $\mathbb{Z}$, and extend it by $v(0) = +\infty$. The following elementary observation is useful when several conditions must hold at the same place.

Lemma 1.1. *Suppose $q_j \neq 0$ and $v(q_j)$ is odd for $0 \leq j < s$. Put*

$$T(\boldsymbol{q}; \boldsymbol{D}) = 1 + \sum_{j=0}^{s-1} q_j^{2^j} D_j^{2^{j+1}}.$$

The finite valuations of the displayed summands are pairwise distinct. Consequently $T \neq 0$ and

$$v(T) = \min\bigl(0,\ 2^j(v(q_j) + 2v(D_j)) : 0 \leq j < s\bigr) \leq 0.$$

If $v(T) = 0$, every nonconstant summand has positive valuation and $T \equiv 1$ in the residue field.

Proof. For a nonzero D_j, the integer $v(q_j) + 2v(D_j)$ is odd. Its product with 2^j is divisible by 2^j and not by 2^{j+1}. Two such integers with different indices cannot coincide, and neither can be zero. The minimum valuation among the nonzero summands is therefore attained exactly once. The ultrametric inequality is an equality in this situation, proving every assertion. Zero values of D_j simply omit their summands. □

We use $K = \mathbb{Q}(\rho)$, $\rho^2 = 2$, and write σ for conjugation. At a prime $p > 2$ the valuations on K above p are unramified and are normalized by $v(p) = 1$.

Proposition 1.2. *Let $R, \delta \in \mathbb{Q}^\times$ and suppose*

$$\nu = v_p(R) \text{ is odd}, \qquad v_p(\delta) = 0.$$

For a positive integer m define

$$q_m = R^{1-2m}\delta^{-2m-2}.$$

Let $W \in K$ be a product of finitely many factors, each of the form

$$1 + \sum_{j=0}^{s-1} q_{m_j}^{2^j} D_j^{2^{j+1}}, \qquad D_j \in K,$$

where each m_j is a positive integer and the orders may vary within a factor and between factors. Zero tests are allowed. Write $W = w_0 + w_1\rho$. Suppose rational numbers T_0, e, b satisfy

$$w_0^2 - 2w_0T_0 - 2w_1^2 = 0, \qquad e\delta b = T_0^2, \qquad e\delta = R^{-1}. \tag{1}$$

Assume also the pole obstruction: $b \neq 0$ implies that $v_p(b)$ cannot be a negative odd integer. Then $\nu > 0$, and there is one place $v \mid p$ at which every test in every factor satisfies

$$v(D_j) \geq m_j.$$

In particular, no inverse witness for w_0 is needed in this assertion.

Proof. For each m, $v(q_m) = (1-2m)\nu$ is odd. Lemma 1.1 therefore gives nonzero factors with nonpositive valuation at every place above p. Thus $W \neq 0$. If $w_0 = 0$, the first equation in (1) gives $w_1 = 0$, a contradiction. Since K is a field, $\mathrm{N}(W) \neq 0$, and the same equation gives

$$T_0 = \frac{\mathrm{N}(W)}{2w_0} = \frac{1}{\mathrm{Tr}(W^{-1})} \neq 0.$$

The second equation now yields

$$b = \frac{R}{\mathrm{Tr}(W^{-1})^2} \neq 0.$$

These equalities have been obtained from the literal polynomial rows; none assumes a deleted inverse condition.

Put $\ell = \max_{v|p} v(W) \leq 0$. The trace inequality gives

$$v_p(\mathrm{Tr}(W^{-1})) \geq -\ell, \qquad v_p(b) \leq \nu + 2\ell.$$

The valuation of b is odd, since it differs from ν by an even integer. The pole obstruction therefore gives

$$0 < v_p(b) \leq \nu + 2\ell. \tag{2}$$

Hence $\nu > 0$ and $\ell > -\nu/2$.

Choose a place attaining ℓ. The valuation of any factor of W is at least ℓ, because all factor valuations are nonpositive and their sum is ℓ. Lemma 1.1 then shows that every nonzero term, whose test is assigned order m_j, has valuation greater than $-\nu/2$. Thus

$$2^j\big((1-2m_j)\nu + 2v(D_j)\big) > -\nu/2,$$

or

$$v(D_j) > \left(m_j - \frac{1}{2} - \frac{1}{2^{j+2}}\right)\nu \geq m_j - \frac{3}{4}.$$

The last inequality uses $m_j \geq 1$ and $\nu \geq 1$. Integrality of the valuation proves $v(D_j) \geq m_j$. The inequality holds trivially when $D_j = 0$. □

2 The rational height detectors

For the curve used in the construction put

$$d = 3 - \rho, \quad D = d^2 = 11 - 6\rho, \quad E : y^2 = x^3 - Dx, \quad f(Q) = 1/x(Q), \quad z(Q) = -x(Q)/y(Q).$$

At the identity both functions extend by zero in the formal neighborhood. The point charts and rational logarithms are those specified in [3, 2]. The logarithms are unchanged here, including their numerator and denominator nonvanishing conditions. Write $U_0 = H$, $V_0 = H - P$, and $Q_i = U_i - V_i$ for the fixed root points. These are elliptic-curve group differences. The exact addition rows and identity-inclusive point charts in the cited specifications enforce these relations; no scalar multiple with a noninteger index is used to define a point.

Each height detector uses a nonzero rational square a, an input $t = a\lambda$, and

$$\delta = t - 1, \qquad \Theta = \left\{0, -\frac{1}{2}, -1, \frac{1}{2399}, -\frac{32}{407}\right\}, \qquad F(t) = \prod_{\theta \in \Theta} (t - \theta).$$

Its contact radical $M(t)$ is the product of primes $p > 76176$ for which $v_p(t - \theta)$ is positive and odd for some $\theta \in \Theta$. The weight equations give

$$F(t)e = (a\delta)^5. \tag{3}$$

On $\delta F(t) \neq 0$ define

$$R = \frac{F(t)}{a^5\delta^6}, \qquad e = \frac{1}{R\delta}, \qquad q_m = \frac{e^{2m-1}}{\delta^3}.$$

The last expression equals the q_m of Proposition 1.2.

Lemma 2.1. *Every solution of (3) has $F(t) \neq 0$. If $p \mid M(t)$, then $\delta \neq 0$, $v_p(\delta) = 0$, and $v_p(R)$ is odd. If $\delta = 0$, then $M(t) = 1$.*

Proof. The five roots in Θ differ from 1. Therefore $F(t) = 0$ contradicts (3), as $a \neq 0$. At a prime greater than 76176, all nonzero differences between distinct elements of $\Theta \cup \{1\}$ are units. A contact prime divides exactly one factor of $F(t)$ to positive odd order and makes δ a unit. Since a is a square, $v_p(F(t)) - 5v_p(a)$ is odd. Finally all prime factors in the reduced numerators and denominators of $1 - \theta$ are at most 76176, so $M(1) = 1$. □

Define the sixteen tests as follows. Their required orders are part of these definitions:

Notation	Test	Order
A_i	$f(U_i)$ $(1 \le i \le 4)$	1
B_i	$\sigma f(V_i)$ $(0 \le i \le 4)$	1
C_1	$L_1 - L_1'$	10
I	$L_1 - \beta$	10
D_H	$f(H)$	40
C_i	$L_i - L_i'$ $(2 \le i \le 4)$	40
G	$L_3^2 - L_2(L_2 - 2)L_3L_4 + (L_2 - 2)^2L_4^2 - 1$	40

Here $\beta = 51X^2 + 2504$, and the L_i, L_i' are the retained primary and alternate ratios, with the latter already conjugated. For an ordered list of tests $(D_0, \ldots, D_{s-1})$, denote by

$$\mathcal{T}(D_0, \ldots, D_{s-1}) = 1 + \sum_{j=0}^{s-1} q_{m(D_j)}^{2^j} D_j^{2^{j+1}} \tag{4}$$

the factor formed with their respective orders.

The two-detector system uses the following six lists:

$$\begin{gathered}(D_H, C_1, B_0), \quad (C_2, I, B_2), \quad (C_3, A_2, B_3),\\ (C_4, A_3, B_4), \quad (G, A_4), \quad (A_1, B_1).\end{gathered} \tag{5}$$

The three-detector system uses these five lists:

$$\begin{gathered}(D_H, C_1, B_1, B_0), \quad (C_2, I, B_2), \quad (C_3, A_2, B_3),\\ (C_4, A_3, B_4), \quad (G, A_4, A_1).\end{gathered} \tag{6}$$

Each of the sixteen tests occurs exactly once in either collection. Every order-forty test occupies position zero. The detector uses the product W of these factors, followed by (1) and the unchanged normalized pole equations. Proposition 1.2 supplies every required local test order at one common place over any contact prime. When $M(t) = 1$, the contact-radical divisibility statements are vacuous; no rational formula for R or W^{-1} is used. This includes $t = 1$.

3 Preservation of simultaneous completeness

The arithmetic hypotheses are those in the specified source construction [1, Section 2]: the elliptic group and index-height assertions, the contact-height inequality, the pole obstruction and pole-realization theorem, and the simultaneous prime-pattern theorem with its quantitative weighted selection hypothesis. The latter allows an arbitrary fixed odd avoidance modulus and the prescribed conjugate reduction orders, fixed finite-place conditions, and real approximation conditions simultaneously. The weight lower bound and slowly decreasing separation parameter are retained exactly as stated there. This section shows that the present replacement requires no stronger assertion.

Proposition 3.1. *For each integer input and each parameter admitted by the scalar construction, the simultaneous point and prime selection in [3, 2] supplies witnesses for the modified height detectors. The local depths and the numerical sufficient scalar parameter are unchanged.*

Proof. First fix the scalar parameter, integer labels, root points, and positive rational square scales by the retained scalar and Lucas construction. Its rational functions are defined on nonempty real and finite-place neighborhoods of the prepared points. Fix compact smaller neighborhoods avoiding their finitely many poles. All test functions occurring in (5)–(6) are bounded on these real neighborhoods. The additional functions are only powers of the same tests; they introduce no new poles or exceptional point exclusions. The continuity modulus and finite avoidance set can be fixed at this stage, before the contact primes are chosen.

The source selection has an exact product description that controls all finite primes, not only the selected contacts. Choose a fixed positive integer D_0 clearing every $D_0^2\theta/a$, and write $v_* = D_0^2 w^2$, $\lambda = u/v_*$. The simultaneous pattern makes each of the five distinct forms $u - (\theta/a)v_*$ a positive selected prime. Consequently

$$R = \frac{\prod_{\theta\in\Theta}(u - (\theta/a)v_*)}{v_*^5\delta^6}. \tag{7}$$

The denominator is a rational square. The selected primes avoid its support and the fixed scale and contact-difference denominators, by the avoidance conditions. Thus $v_p(R) = 1$ at the five selected contacts of this detector and $v_p(R)$ is even at every other prime. This is the displayed product formula in the retained comparison construction's completeness proof in [6]; it also follows directly by substituting $t = au/v_*$ in the definition of R.

At a selected contact prime the construction has $v_p(R) = 1$ and a designated primary place at which each original order-m test satisfies $v(D_j) \geq m$. Thus $q_m D_j^2$ has positive valuation

there. Every nonconstant term of a modified factor is a positive integer power of one of the expressions $q_{m_j} D_j^2$. Consequently each modified factor is 1 modulo the primary prime. Thus $W \equiv 1$ there.

At the opposite split place, Lemma 1.1 shows that each modified factor either has negative valuation or is 1 modulo the prime. If at least one factor has negative valuation, then W^{-1} is zero modulo that prime; otherwise $W^{-1} \equiv 1$. The two contributions to $\mathrm{Tr}(W^{-1})$ therefore sum to 1 or 2 modulo p. Since $p > 76176 > 2$, the trace is a unit. Thus the pole input $b = R/\,\mathrm{Tr}(W^{-1})^2$ has the required positive odd valuation at every selected contact prime. The selection excludes the fixed denominators of the root data and the scales, as in the source.

At the finitely many prepared places, the source preparation makes each $q_m D_j^2$ as small as needed. Taking its second, fourth, or eighth power preserves and strengthens this property. Hence the modified W is arbitrarily close to 1 there. The required local squareclasses and valuation bounds for b at $2, 3, 17$ persist after increasing the same finite preparation depths. At every prime outside the selected contacts, (7) gives an even valuation of R, and hence an even valuation of $b = R\,\mathrm{Tr}(W^{-1})^{-2}$, regardless of its sign. At the selected contacts the preceding unit argument gives valuation one. These two assertions together prove that the reduced denominator of b is a square. Moreover each selected prime is square in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$ by its prescribed congruences, so (7) gives the required local squareclasses.

For real positivity and size, let λ tend to positive infinity after the root data and scales have been fixed. Uniformly on the chosen compact real neighborhoods,

$$R = \frac{1 + O(\lambda^{-1})}{a^6 \lambda} > 0, \qquad q_m = a^{10m-8} \lambda^{-3} (1 + O(\lambda^{-1})).$$

All new factors are positive at both real embeddings, because every $q_m > 0$ and all powers of the tests are even. They tend uniformly to 1; a term in position j decays as $O(\lambda^{-3 \cdot 2^j})$. Therefore

$$\mathrm{Tr}(W^{-1}) \longrightarrow 2, \qquad b = \frac{R}{\mathrm{Tr}(W^{-1})^2} \longrightarrow 0^+.$$

For sufficiently large λ, the pole input lies in $(0, 17)$. The same weighted selection premise supplies these arbitrarily large parameters after the fixed congruences and avoidance conditions; its decay requirement has not increased. Applying the pole-realization input gives fresh simultaneous pole witnesses.

All remaining circuit coordinates are filled by their rational definitions. In particular no assertion is made that an arbitrary old pole witness is preserved. The numerical lower bound on the fixed scalar parameter is used solely in soundness, where the test orders and height inequalities are unchanged. Thus it is not enlarged by this completeness argument. □

4 A quadratic implementation using existing powers

The variable $w = Df^2$ is used in the rational logarithm at each V_i. For $i = 0, 2, 3, 4$, that part of the circuit already contains w^2 and w^4. At V_1 it already contains w^2. These coordinates are still required by the unchanged logarithm approximation.

The stored normalization is important. If u, v are the two stored coordinates for w^2, then the actual value is

$$w^2 = 193u - 132v\rho.$$

Since $D^2 = 193 - 132\rho$ and $\mathrm{N}(D^2) = 2401$, one has

$$f^4 = \frac{(37249u - 34848v) + (25476u - 25476v)\rho}{2401}. \tag{8}$$

If r, s are the stored coordinates for w^4, its actual value is $r + s\rho$. As $D^4 = 72097 - 50952\rho$ and $\mathrm{N}(D^4) = 5764801$,

$$f^8 = \frac{(72097r + 101904s) + (50952r + 72097s)\rho}{5764801}. \tag{9}$$

Conjugating the right sides gives the expressions consumed in (5)–(6). Thus all powers of the V_i needed there are linear forms in existing coordinates.

For (5), exactly five additional field powers are needed: $C_1^4, I^4, A_2^4, A_3^4, A_4^4$. Each is the square of its existing square. All four eighth powers $B_0^8, B_2^8, B_3^8, B_4^8$ are already available from (9); B_1^4 is available from (8).

For (6), exactly nine additional field powers are needed:

$$C_1^4, I^4, A_2^4, A_3^4, A_4^4, \quad B_1^8, B_0^{16}, \quad A_1^4, A_1^8.$$

The sixteenth power B_0^{16} is the square of the existing eighth power, and B_1^8 is the square of the existing fourth power. Every listed field square uses two rational coordinates and two quadratic equations. Explicitly, if $u + v\rho$ is a linear form in available coordinates, its square is represented by

$$j_0 = u^2 + 2v^2, \qquad j_1 = 2uv.$$

These equations have unique rational solutions for their output coordinates, including when the field value is zero.

The affine normalizations of the input and Lucas squares are retained. In source coordinates, they are

$$I^2 = (x_{307} - 6x_{232} + 306x_{248} + 15015) + (2x_{308} - 6x_{233})\rho,$$
$$G^2 = (x_{319} - 2x_{253} + 1) + (4x_{320} - 4x_{254})\rho.$$

The independent checker derives the same expressions and verifies that the retained square rows have these interpretations.

The weight circuit stores $q = q_1$, $q_2 = q^2$, and $q_4 = q_2^2$. Its retained quadratic identities on the nonzero domain are

$$a_1 = q/e, \quad b_1 = a_1^2/e, \quad c_1 = b_1^2/q, \quad c_1 d_1 = q q_4, \quad c_1 q_{10} = d_1^2, \quad c_1 q_{40} = q_{10}^4.$$

Since $q = e/\delta^3$, these yield

$$c_1 = e^{-3}\delta^{-9}, \quad d_1 = e^8\delta^{-6}, \quad q_{10} = e^{19}\delta^{-3}, \quad q_{40} = e^{79}\delta^{-3}.$$

The square of q_{10} is already stored for the last identity. It also supplies the order-ten terms in position one. Every order-forty test occupies position zero, so no square or cube of q_{40} is required. For (6), add one scalar coordinate per detector for $q_8 = q_4^2$; this supplies the position-three term. For (5) no such coordinate is required. All weight rows are quadratic. At a consumed contact prime their nonzero interpretation follows from Lemma 2.1; no division is asserted at an empty contact radical.

These choices give the following exact coordinate accounting relative to the specified systems with the same height scales:

Contribution	Two detectors	Three detectors
Starting coordinates	317	370
Shared expanded coefficients removed	−22	−22
New field powers	+10	+18
Squares and cubes of q_{40} removed	−4	−6
Eighth powers of q_1	0	+3
Factor and product coordinates removed	0	−12
Total	301	351

There is a corresponding change in the number of quadratic rows. The literal systems contain 275 and 323 rows respectively. All remaining rows, including the cleared trace and the scalar guard, are retained without alteration.

5 Specified quartics and verification

The exact tables specify primitive integer quadratic rows; their squared sum is the defining polynomial. Every count includes the input coordinate. The parameter $A = 80989^{2N+1}$ is a fixed integer coefficient, not a polynomial variable.

Family	Total variables	Quadratic rows	Ordinary monomials
Two height scales	301	275	4305
Three height scales	351	323	4990

The ordinary monomial counts hold for $N \geq 27464$. Before terms with different powers of the fixed parameter are collected, the tables have 4320 and 5005 terms respectively. Every polynomial has degree exactly four and content one. Both coefficient heights satisfy

$$H(P_N) < 2^{1520} 80989^{4N+2}.$$

The 351-variable member with $N = 27464$ has a largest coefficient with 1791357 binary digits. The largest coefficient is its constant term.

The producer `build_quartic_mixed.py` constructs both complete systems and coefficient tables from the pinned source systems. The separate program `check_quartic_mixed.py` imports no producer. It reconstructs the field constants from $D = 11 - 6\rho$, verifies every changed and unchanged quadratic row, and independently expands every square using unordered pairs of terms. It checks the entire coefficient table, the coordinate incidence and counts, the degree and content, and the exact height of the explicit member. Its finite valuation checks supplement Proposition 1.2; the proposition proves the assertion for all odd valuations, without a search cutoff.

For completeness, the coefficient-height bound is a direct integer certificate. For each coordinate monomial, write its coefficient as $c_0 + c_1 A + c_2 A^2$. The maximum of $|c_0| + |c_1| + |c_2|$ has 1520 binary digits. Since $A \geq 1$, the displayed bound follows. For $N \geq 27464$, $A > 2^{1520} + 1$, so the highest nonzero term of each such coefficient dominates the sum of the lower terms. This also verifies that no ordinary monomial disappears at an allowed parameter.

The conditional projection statements and their different effectiveness properties are stated separately in [4, 5]. Finite algebraic verification establishes none of the deep arithmetic premises listed in Section 3.

References

[1] *A forced prime condition and an explicit quartic in 392 variables*, 8 November 2026, Section 2; accompanying source table and independent arithmetic certificates. PDF; source: `quartic392_20261108_v8.tex`.

[2] *A family of quartics in 317 variables from aggregate trace heights*, 8 November 2026, exact family and conditional projection theorem. PDF; source: `quartic317_family_20261108_v9.tex`.

[3] *A quartic in 370 variables with specified integer coefficients*, 8 November 2026, exact family and simultaneous completeness proof. PDF; source: `quartic370_20261108_v10.tex`.

[4] *A family of quartics in 301 variables*, 9 November 2026, accompanying theorem and coefficient tables. PDF; source: `quartic301_20261109_v11.tex`.

[5] *An explicit quartic in 351 variables*, 9 November 2026, accompanying theorem and coefficient tables. PDF; source: `quartic351_20261109_v11.tex`.

[6] E. Glazer, *An explicit quartic defining the integers in the rationals*, Version 13, manuscript and coefficient data, commit `5bbd74f6a479776a332cbe036809d5dddae63720`, https://github.com/elliotglazer/diophantine-z-q.